INSPECTION REPORT

ChArUco target · ceramic · 200 × 150 mm

ITEM

Part number	CT-CHA-C-200150-6X6
Description	ChArUco calibration target, ceramic, 200 × 150 mm
Serial number	000311
Lot	L-2611-0382
Datasheet	CT-DS-CHA-C · Rev A
Pattern	ChArUco 12 × 17 · 10 / 7 mm DICT_6X6_250
Marker ID range	0–101 (102 markers)
Start ID	0

INSPECTION

Inspection date	2026-10-24
Inspector	JG
Location	per Calib.Tools procedure CT-WI-001
Instrument	Optical CMM · OCMM-01
Last calibration	2026-03 · valid to 2027-03
Temperature	20.4 °C ambient · 20 °C reference

OVERALL RESULT

PASS

All measured features within datasheet tolerance.

9 geometry · 1 flatness · 4 visual
176 corners · U = 1.5 µm (k = 2)

1 Method

Positions of all 176 ChArUco corners were measured on an optical CMM (OCMM-01); flatness was measured at 25 points. Expanded uncertainty U = 1.5 µm (k = 2). Results are corrected to the reference temperature of 20 °C; ambient temperature during measurement 20.4 °C.

2 Pattern geometry

ceramic: deviations in µm

FEATURE	NOMINAL (mm)	MEASURED (mm)	DEVIATION (µm)	TOLERANCE (µm)	RESULT
Square pitch X	10.000	10.0012	+1.2	±2	PASS
Square pitch Y	10.000	9.9992	−0.8	±2	PASS
Pattern width	170.000	170.004	+4	±5	PASS
Pattern height	120.000	120.003	+3	±5	PASS
Diagonal 1	208.087	208.091	+4	±7	PASS
Diagonal 2	208.087	208.085	−2	±7	PASS
Orthogonality	90.000°	90.0008°	–	±0.002°	PASS
Max single-feature deviation · n = 176	–	–	1.6	≤ 2	PASS
RMS deviation · information only	–	–	0.7	–	N/A

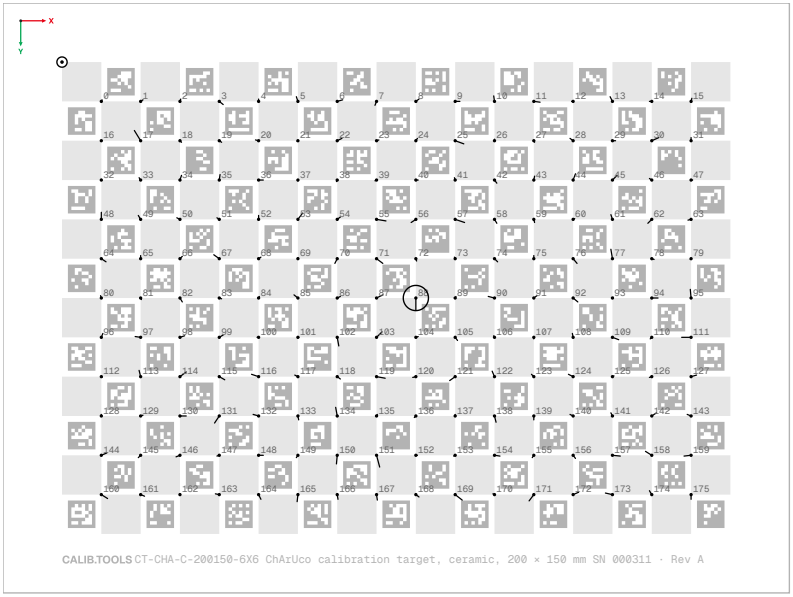
3 Flatness

MEASURED FLATNESS (µm)	TOLERANCE (µm)	METHOD	RESULT
12	≤ 20	25 points, 5 × 5 grid, least-squares plane	PASS

4 Visual

CHECK	CRITERION	RESULT
Edge chamfer	0.5 × 45° on all edges, no chips > 0.3 mm	PASS
Surface defects	No scratches or pinholes in the pattern area	PASS
Print / edge definition	Square edges sharp, no ink bleed at 10× loupe	PASS
Marking legibility	Label line, X / Y arrows and SN legible	PASS

5 Feature map



LEGEND

X → right

Y ↓ down

⊙ Origin (0, 0): top-left outer corner of the pattern (OpenCV frame)

• Measured corner with id, n = 176

—— Deviation vector, magnified × 2000 (1 µm = 2 mm on the board)

◯ Largest deviation: id 88, 1.6 µm

6 Measured coordinates

Complete corner coordinates are supplied in CT-CSV-000311.csv (176 rows; columns id, row, col, x_nom, y_nom, x_meas, y_meas, dx_um, dy_um; mm unless stated). First 10 rows:

id	row	col	x_nom	y_nom	x_meas	y_meas	dx_um	dy_um
0	0	0	10.000	10.000	10.0001	9.9998	+0.1	-0.2
1	0	1	20.000	10.000	20.0003	9.9998	+0.3	-0.2
2	0	2	30.000	10.000	30.0001	9.9997	+0.1	-0.3
3	0	3	40.000	10.000	40.0005	10.0004	+0.5	+0.4
4	0	4	50.000	10.000	50.0000	9.9997	0.0	-0.3
5	0	5	60.000	10.000	59.9998	9.9994	-0.2	-0.6
6	0	6	70.000	10.000	70.0007	9.9998	+0.7	-0.2
7	0	7	80.000	10.000	79.9998	10.0005	-0.2	+0.5
8	0	8	90.000	10.000	90.0004	9.9997	+0.4	-0.3
9	0	9	100.000	10.000	100.0006	10.0000	+0.6	0.0

7 Statement

The item identified above meets the specifications of datasheet CT-DS-CHA-C rev A at the time of inspection. This is an inspection report issued by Calib.Tools; it is not an accredited calibration certificate. An ILAC-MRA certificate issued by an accredited third-party laboratory is available as SVC-X-CERT-ILAC.

8 Traceability

Instrument	OCMM-01 · certificate K-2026-03-0141 · valid to 2027-03
Reference standard	Golden sample GS-CHA-C-200150-6X6 · DAkkS certificate [number] · valid to [date]
Raw data	Raw data retained for 10 years under this serial number.

SIGNED

JG

JG · Inspector · 2026-10-24
Printed signature

CALIB.TOOLS SN 000311 INSPECTED 2026-10-24 JG DIMENSIONAL INSPECTION

calib.tools/verify/000311